ASPIRATION DISSIMILATION AND PHONOLOGICAL RULES IN TANGKHUL NA

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Journal/Venue: Carnegie Mellon University Working Papers in Linguistics | Year: 2016
Subject: Phonetics & Laboratory Phonology | Language: English & Tangkhul

ABSTRACT:

In Tangkhul Naga, prefixal stop consonants exhibit systematic aspiration dissimilation reminiscent of Grassmann's Law in Indo-European. When a verbal root begins with an aspirated stop (ph-, th-, kh-), the preceding nominalizing/infinitive prefix surfaces as an unaspirated stop [ke-]. Conversely, when the verbal root begins with an unaspirated stop or sonorant, the prefix is realized with robust aspiration [khe-] in conservative varieties. Using acoustic measurements of Voice Onset Time (VOT) and burst spectra from 8 native speakers, this paper confirms that aspiration is a syntagmatically governed feature bounded by the phonological word.

Keywords: Aspiration, Dissimilation, Grassmann's Law, Phonology, VOT, Acoustics

1. THE PHENOMENON OF ASPIRATION DISSIMILATION

Phonologists have long studied long-distance consonant interactions such as consonant harmony and dissimilation. In Tangkhul, the interaction between the prefix consonant and the root-initial consonant exhibits a classic Obligatory Contour Principle (OCP) effect for the feature [spread glottis]. Two adjacent syllables within the same phonological stem cannot both surface with contrastive aspiration.

Tangkhul Linguistic Data & Analysis:

*** ke-pha [kapha] : good, beautiful**

Analysis: Root initial aspirated [ph-]; prefix is unaspirated [ke-].

*** khe-pa [khapa] : to read / recite**

Analysis: Root initial unaspirated [p-]; prefix surfaces with aspiration [khe-].

*** ke-thi [kathi] : to die**

Analysis: Root [th-]; prefix dissimilates to unaspirated stop.

2. ACOUSTIC METHODOLOGY & VOICE ONSET TIME (VOT)

Acoustic recordings of 120 prefix-root words were analyzed in Praat. Burst duration, Voice Onset Time (VOT), and vowel fundamental frequency (F0) perturbation were quantified. Aspirated root onsets showed mean VOTs of 68-85 ms, while unaspirated root onsets averaged 12-22 ms. The prefix vowel duration showed compensatory shortening when preceding aspirated roots, indicating a prosodic foot-level timing constraint.

Tangkhul Linguistic Data & Analysis:

*** khe-sa [khasa] (to be hot) : VOT = 18 ms (root); Prefix aspiration = 45 ms**

Analysis: Prefix aspiration licensed before fricatives.

*** ke-kham [kakham] (to stop/hinder) : VOT = 78 ms (root); Prefix aspiration = 0 ms**

Analysis: Strict OCP dissimilation.

3. THEORETICAL PHONOLOGICAL IMPLICATIONS

These findings demonstrate that Tangkhul prefixal alternation is not mere morphological suppletion, but an active phonological rule governed by laryngeal licensing. It provides one of the few well-documented non-Indo-European cases of Grassmann's Law in the Sino-Tibetan language family.

Tangkhul Linguistic Data & Analysis:

*** ke-that [kathat] : to kill**

Analysis: PTB *g-sat; aspiration conditioned by ancient sibilant prefix.

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